Course Code: BAI405E | **Course: Linear Algebra and Optimization**
Credits: 3 | **L:T:P:S 3:0:0**
SEE: 50% Marks | **CIE: 50% Marks**
SEE Hours: 3 Hrs | **Max. Marks: 100**

Professional Elective for (IV Sem CSE / ISE / AI&ML)

Prerequisites if any	Basic Linear Algebra, Calculus
Learning objectives	The goal of the course **Linear Algebra and Optimization is to** 1. To develop familiarity with computational techniques of Linear Algebra for dimensionality reduction. 2. Familiarize with the concepts of both constrained and unconstrained optimization.

Course Outcomes:

On the successful completion of the course, the student will be able to

Course Outcomes		**Bloom's level**
CO1	Apply SVD's for dimensionality reduction	Understand, Apply, Analyze
CO2	Apply the concepts of Calculus and Linear Algebra for Optimization.	

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3	PSO4
CO1	3	2	-	1	-	-	-	-	-	-	-	1		-	-	-	-
CO2	3	2	-	1	-	-	-	-	-	-	-	1		-	-	-	-

Mapping Strength: Strong– 3 Medium – 2 Low – 1

ESTD : 1946

Course Structure

		No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
	Module – 1			
1.1	Basic review of Linear Algebra	3	-	-
1.2	Positive definiteness	1	-	-
1.3	Quadratic forms	2	-	-
1.4	Singular value decomposition.	2	-	-
	Module – 2			
2.1	Least square solutions.	2	-	-
2.2	Applications to Linear models	4	-	-
2.3	Principal component analysis.	2	-	-
	Module – 3			
3.1	Linear optimization: The graphical method.	1	-	-
3.2	Simplex method.	2	-	-
3.3	Integer programing problems-branch and bound method.	3	-	-
3.4	Duality, degeneracy.	2	-	-
	Module – 4			
4.1	Basic review of vector calculus.	2	-	-
4.2	Non-linear Unconstrained Optimization problems.	2	-	-
4.3	Newton's method.	2	-	-
4.4	Descent methods.	2	-	-
	Module – 5			
5.1	The derivative matrix.	2	-	-
5.2	Level sets and level surfaces.	1	-	-
5.3	Non-Linear constrained optimization problems- Lagrange multiplier method	3	-	-
5.4	Applications of Quadratic forms in constrained optimization.	2	-	-
Total No. of Lecture Hours		**40**	-	-

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Detailed Lesson Plan:

Sr No. of Module	Number of related learning Objectives	Weeks/ Dates	Online Mode		ICT Tool/ Platform/ LMS	Face-to-face Mode	
			Resource (OER/ URL/ IM/ CP)	Activity (Describe activity in detail)		Resource (OER/ URL/ IM/ CP)	Activity
1.1	1		Essential Mathematics for Machine Learning	-		-	Group Discussion & Presentation
1.2	1			-		-	
1.3	1			-		-	
1.4	1			-		-	
2.1	1			-		-	
2.2	1			-		-	
2.3	1			-		-	
3.1	2			-		-	
3.2	2			-		-	
3.3	2			-		-	
3.4	2			-		-	
4.1	2			-		-	
4.2	2			-		-	
4.3	2			-		-	
4.4	2			-		-	
5.1	2						
5.2	2						
5.3	2						
5.4	2						

Assessment Pattern:

Bloom's level	Test1	Test2	Quiz	SEE
Remember	✓	✓	✓	✓
Understand	✓	✓	✓	✓
Apply	✓	✓	✓	✓
Analyze	✓	✓		✓

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Reference Books:

1. Gilbert Strang, Introduction to linear algebra, Wellesley Cambridge press, 4th edition.
2. Chong and Zak, An introduction to optimization, Wiley, second edition.
3. David C Lay, Linear Algebra and its Applications, Pearson, Fifth edition.